Assignment : Internet Explorer 09/02

Client - Server Architechture : What happens when we open a website.

Client : device / software that requests service.
Responsible to send request, display response, handle user interaction.

Server : powerful computer that provides services / resources to client.
Responsible to recieve request, process it, communicate with database.

Step - by - Step Process :

Step1 : User enter website URL
ex : www.youtube.com

Step2 : Browser contacts DNS Server to convert the domain name into IP address, to identify server location.
ex: www.youtube.com → 192.168.1.1

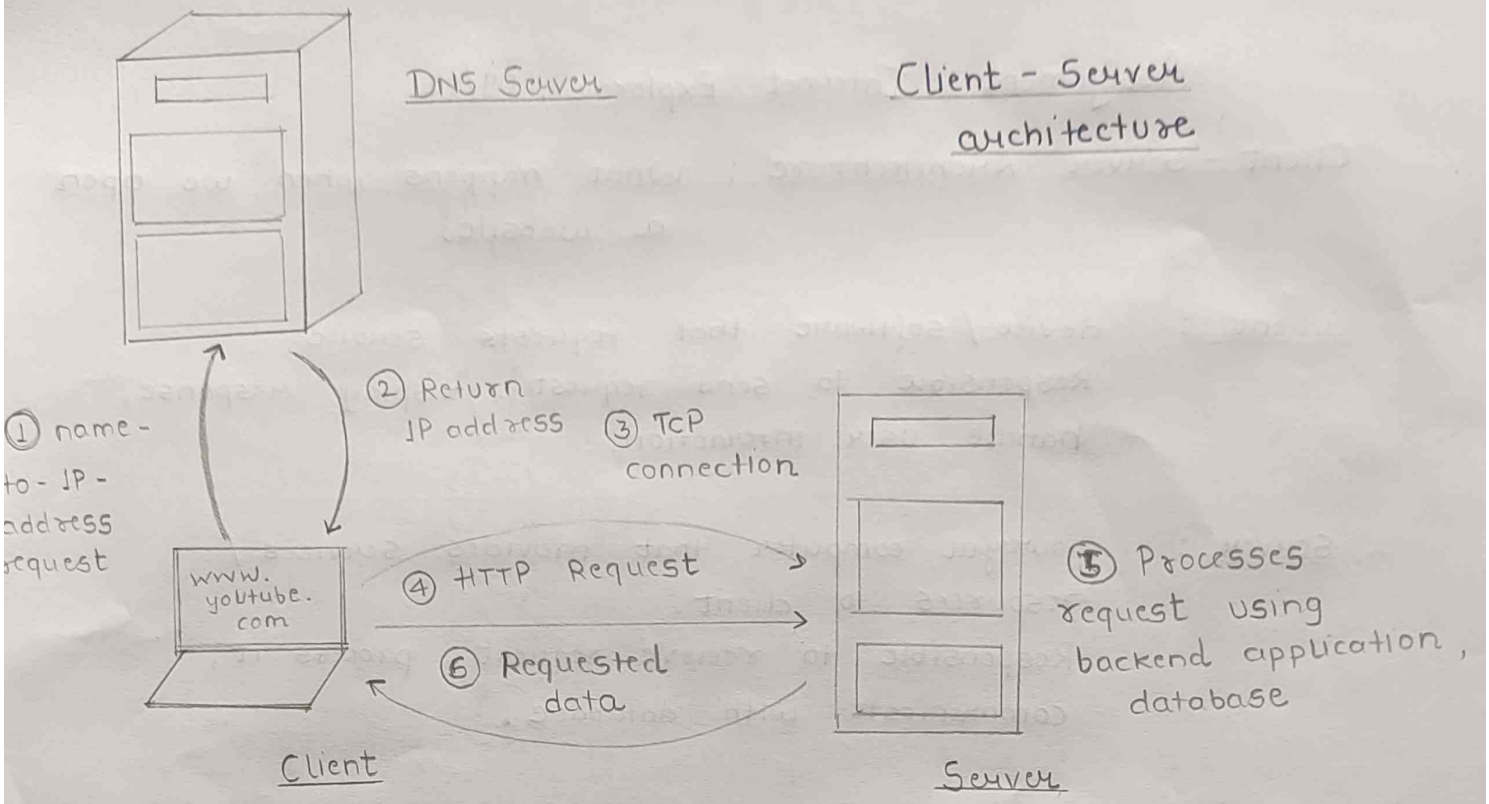
Step3 : Browser establishes TCP connection with server

Step 4 : Browser sends HTTP request to server
Ex : GET or POST

Step5 : Server receives the request, sends to backend application, which in turn may use database & then requested data is returned.

Step 6 : Server sends back ; status code, requested data, files etc.

Step 7 : Browser renders the data received, displays the data.


DNS Server
Client - Server architecture
① name-to-IP-address request
② Return IP address
③ TCP connection
www.youtube.com
④ HTTP Request
⑤ Processes request using backend application, database
⑥ Requested data
Client
Server